

Near-duplicate leakage can reorder model rankings on a genomic benchmark suite: an audit of Genomic Benchmarks

Rikhin Kavuru^{1,*}

¹Independent Researcher

*Corresponding author. rikhinkavuru@gmail.com

Abstract

Motivation: Benchmarks for DNA sequence classification underpin method comparison, but standard train/test splits rarely control for near-duplicate sequences that span the split. When near-identical sequences appear on both sides, a model can score well by memorizing near-duplicates rather than generalizing. Whether this merely inflates scores, or actually changes *which* method appears best, has not been systematically characterized on a widely used suite. **Results:** We audit the seven binary datasets of the Genomic Benchmarks suite, quantifying train/test overlap by exact and near-duplicate k -mer detection and re-evaluating four classifiers under a near-duplicate-aware re-split. Two datasets are clearly leaky and four are clean under both a k -mer and a length-robust containment metric; a fifth is borderline, flagged only by the length-robust metric. Correcting the split lowers the best model's (a default random forest) accuracy by 11.8 points on *human_nontata_promoters* ($0.928 \rightarrow 0.810$) and 16.5 points on *human_enhancers_ensembl* ($0.860 \rightarrow 0.695$); cluster (block) bootstrap CIs exclude zero, and all clean-dataset deltas include zero. Most consequentially, leakage can reorder the ranking. On *human_enhancers_ensembl* the best model flips (random forest $\rightarrow$ *LinearSVC*); this is airtight—it holds under accuracy, AUROC and F_1 , reproduces under a chromosome-holdout control, and the bootstrap probability that the random forest ranks first goes from 1.0 to 0.0. On *human_nontata_promoters* the demotion appears only under accuracy at threshold 0.5 and survives neither a threshold-free metric nor the chromosome control, so the ranking-change claim is demonstrated cleanly on *one* dataset and is partial on the other. The mechanism is memorization of label-carrying near-duplicates by the default unregularized random forest: regularization removes both the memorization and the apparent lead. A from-scratch 1D convolutional network shows the same inflation (the regularized-CNN drop CI excludes zero on both datasets), so it is not a classical-model artifact, and alignment identity (MMseqs2) confirms the *human_enhancers_ensembl* effect metric-independently. We provide a per-dataset report card and a dependency-free, near-duplicate-aware splitter. **Availability and implementation:** Code and data are available at <https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks>; the classical audit is CPU-only and fully reproducible, and the deep-model checks run deterministically on a single GPU/MPS device.

Keywords data leakage, near-duplicate sequences, benchmark evaluation, genomics, model ranking

1. Introduction

Machine learning for DNA sequence classification is now routine, and benchmark suites have become the default proving grounds for new methods. The Genomic Benchmarks collection (Grešová et al., 2023) is widely used precisely because it is small, accessible, and quick to run, making it a natural first stop when comparing models. The scientific value of any such suite, however, rests entirely on one assumption: that its train/test split measures generalization rather than recall of the training data.

Near-duplicate leakage threatens this assumption. We use *near-duplicate leakage* to mean the situation in which exact or near-identical sequences appear on both sides of a split; where local-alignment or containment evidence earns the stronger word, we call the shared-ancestry variant *homology*. A model

can then succeed on the test set by recognizing memorized near-copies of training examples rather than by learning generalizable signal, so reported accuracy overstates true generalization. This failure mode is increasingly documented across biological machine learning, from regulatory-genomics regression to protein-interaction prediction and virtual screening (Section 2).

Two claims must be kept separate. **Claim I**—that de-leaking a split lowers apparent accuracy, i.e. that leakage inflates scores—is not our contribution: it is already established across domains by prior work, including the general leakage-and-reproducibility analysis of Kapoor and Narayanan (2023), the long-range genomic regression study of Rafi et al. (2025), and the protein-interaction benchmarks of Bushuev et al. (2024). We reproduce it here only as a within-suite control.

Claim II—that near-duplicate leakage can *reorder* which model

appears best—is our contribution, and it is scoped: it requires both short, fixed- or near-fixed-length human regulatory DNA and a memorization-prone model in the comparison set. We demonstrate Claim II cleanly on one dataset and, honestly, only partially on a second.

Our contributions are:

- a per-dataset near-duplicate *report card* for the binary suite with a matched clean-versus-leaky control, showing that leakage is dataset-specific rather than universal, and that a length-robust containment re-measure both confirms four clean verdicts and flags one borderline dataset the length-blind metric misses;
- evidence that near-duplicate leakage can *reorder the model ranking* when a memorization-prone model is in the comparison set—airtight on *human_enhancers_ensembl* (accuracy, AUROC, F_1 , a chromosome-holdout control, and a bootstrap winner probability all agree) and a cautionary partial case on *human_nontata_promoters*;
- a mechanism, validated three ways: a graded memorization analysis plus a random-forest regularization path (the effect is a property of the *default* unregularized forest), a from-scratch 1D CNN dose-response (the effect is not a classical-model artifact), and a chromosome-holdout and alignment-based re-measure (the effect is not a clustering or k -mer artifact); and
- a dependency-free, near-duplicate-aware splitter, with an honest statement of scope.

The classical audit is deterministic, CPU-only, and reproducible; the deep-model checks are pre-registered and run deterministically on a single device.

2. Related work

Leakage is an established benchmarking hazard. Kapoor and Narayanan (2023) document leakage as a pervasive driver of the reproducibility crisis in ML-based science and show that it can inflate reported performance and, in extreme cases, change the apparent ordering of methods; our study is a genomics-specific, controlled instance of that phenomenon. In genomics, hashFrag (Rafi et al., 2025) shows that homologous sequences spanning genomic train/test splits inflate the apparent performance of neural models—test performance scales with similarity to the training set—and provides tools to detect and partition such data. In protein–protein interaction prediction, Bushuiev et al. (2024) show that standard splits leave most test cases with near-duplicate counterparts, rewarding overfitting capacity rather than generalization; PINDER (Kovtun et al., 2024) similarly shows that de-leaked test sets sharply reduce reported performance, including for AlphaFold-Multimer. In virtual screening, an audit of LIT-PCBA (Huang et al., 2025) shows that a trivial memorization baseline can match sophisticated methods once leakage is accounted for.

The closest methodological prior art is homology-aware partitioning. GraphPart (Teufel et al., 2023) partitions biological sequences so that no train/test pair exceeds a similarity threshold, and CD-HIT (Li and Godzik, 2006) and MMseqs2 (Steinegger and Söding, 2017) are the established clustering engines for this task. Our near-duplicate-aware splitter is a

lightweight, dependency-free instance of the same whole-cluster idea; we validate against MMseqs2 alignment identity to confirm the effect is not an artifact of our k -mer edge metric.

We differ from prior genomics work in what we measure. Those studies document inflated absolute scores within a single leaky domain; we audit a whole suite with a within-study clean control and ask the downstream question—whether leakage reorders *which model wins*. We also position our audit against the deep-model literature it is often compared to: DNABERT-2 (Zhou et al., 2023), HyenaDNA (Nguyen et al., 2023), and the Nucleotide Transformer (Dalla-Torre et al., 2024) are pretrained on the same human reference genome from which these datasets are drawn, so evaluating them here would introduce a second, pretraining-based leakage channel that no train/test split can close (Section 5); we therefore use a from-scratch CNN as the deep-model probe and scope the pretrained foundation models out of the split-effect evidence.

3. Methods

3.1. Datasets

We study the seven binary datasets of the Genomic Benchmarks suite (Grešová et al., 2023), and additionally report the three-class *human_ensembl_regulatory* set separately. Per-dataset sizes, sequence lengths, and class balance are given in Table 1. Sizes range from 6,914 (*drosophila_enhancers_stark*) to 174,756 (*human_ocr_ensembl*) sequences; lengths are fixed within some datasets (e.g. 251 bp for *human_nontata_promoters*) and variable in others; classes are balanced (0.50 positive fraction) except *human_nontata_promoters* (0.544). All balanced datasets are curated to 0.500 in both train and test (positive/negative subfolders), so class balance is preprocessed, not natural—which is why we also run an imbalanced-prevalence stress test (§3.10).

3.2. Features, models, and decision rule

Each sequence is represented by overlapping k -mer counts for $k \in \{4, 6\}$ (4^k features). We evaluate four classifiers spanning a range of *memorization propensity* (equivalently, effective regularization) rather than raw capacity: logistic regression ($C=1.0$, $\text{max_iter}=5000$), a linear support-vector machine (LinearSVC, $C=1.0$, $\text{dual}=\text{False}$), a random forest (150 trees, scikit-learn defaults $\text{min_samples_leaf}=1$, $\text{max_depth}=\text{None}$), and gradient-boosted trees (HistGradientBoosting, library defaults). L2 normalization is applied *only* to the two linear models, via a Normalizer pipeline step; the tree ensembles (random forest, HistGradientBoosting) consume raw counts, for which monotone feature scaling is irrelevant. We deliberately avoid a “capacity” framing: the nominally higher-capacity HistGradientBoosting is barely inflated while the random forest is heavily inflated, so the controlled variable is memorization propensity, not capacity. Unless noted, all accuracies use a single decision rule—`.predict()` (`argmax` of the class probabilities)—and the headline accuracy drop is the difference between the original accuracy and the mean corrected accuracy over five re-split seeds, with a percentile test-set bootstrap 95% CI (1,000 resamples of the per-example correctness vector; models are

not refit). To confirm strand-symmetry is not the driver, we also re-run the full comparison with canonical k -mer features, $\min(k\text{-mer, reverse complement})$.

3.3. Near-duplicate measurement

We measure train/test overlap by the exact Jaccard similarity of k -mer sets ($k = 8$), computed for all test–train pairs via sparse binary matrix products (no MinHash approximation). For each test sequence we record its maximum similarity to any training sequence, and define the leakage fraction as the proportion of test sequences whose maximum similarity exceeds a threshold (0.5, 0.7, 0.9). Because a k -mer-set Jaccard is bounded by the length ratio ($\text{Jaccard} \leq L_{\min}/L_{\max}$), it undercounts leakage on datasets with variable or disparate lengths; we therefore also report a length-robust *containment* index, $|A \cap B|/\min(|A|, |B|)$, in the same clustering, and disclose the length-pair cap per dataset (Table 2). Jaccard leak fractions are thus lower bounds; the containment re-measure is the length-robust check on the clean verdicts. We additionally census exact byte-identical train/test duplicates: on *human_enhancers_ensembl*, 11,774 test sequences (0.380 of the test set) are byte-identical training copies (leakage is literal duplication), whereas *human_nontata_promoters* has 0 exact duplicates yet 0.225 near-duplicates at Jaccard ≥ 0.9 (genuinely mutated copies)—the more homology-like but statistically weaker case.

3.4. Near-duplicate-aware re-split

To correct leakage we build a similarity graph in which sequences are connected when their k -mer Jaccard exceeds 0.7, take connected components as near-duplicate clusters, and assign whole clusters to either train or test. The assignment preserves the original train/test ratio and class balance and, by construction, guarantees zero residual cross-split similarity above the threshold (verified empirically). We confirm robustness to the clustering threshold (sweeping 0.5/0.7/0.9) and to removal of the single largest component. This is a dependency-free instance of the whole-cluster partitioning used by GraphPart (Teufel et al., 2023), CD-HIT (Li and Godzik, 2006), and MMseqs2 (Steinegger and Söding, 2017), any of which could substitute for the edge metric.

3.5. Alignment-based validation

To test whether the effect depends on the k -mer edge metric, we re-cluster the two leaky datasets with MMseqs2 (Steinegger and Söding, 2017) *easy-cluster* (*min-seq-id* 0.7, *-c* 0.8), feed the external cluster vector into the *same* whole-cluster assignment path, and refit all four models—only the edge metric changes (alignment identity vs 8-mer Jaccard), the linkage held fixed. Because MMseqs2 is itself k -mer-seeded, we describe this as an *alignment-scored* cross-check, not a k -mer-independent one.

3.6. Leave-one-chromosome-out control

We recover source genomic coordinates for the two leaky datasets and build a chromosome-holdout split—whole chromosomes are assigned entirely to the test

side until the target test fraction is reached (nine chromosomes for *human_enhancers_ensembl*, twelve for *human_nontata_promoters*)—the field-standard control in regulatory genomics, refitting all four models. This control is deployment-relevant but *not* leakage-free: residual near-duplicate leakage surviving the chromosome split is 0.8% (*human_enhancers_ensembl*) and 0.02% (*human_nontata_promoters*) at Jaccard 0.7 (versus the exact 0 guaranteed by the near-duplicate-aware split), which we report rather than hide.

3.7. Regularization path and the graded memorization gap

To test whether the effect is an artifact of unregularized random-forest defaults, we sweep *min_samples_leaf* $\in \{1, 5, 20, 50, 100, 200, 500\}$ and *max_depth* $\in \{\text{None}, 16, 8, 4\}$ at full scale (the largest settings exceed the largest near-duplicate cluster, so no single leaf can be a pure duplicate block). The confound-immune headline metric is the split-internal *graded memorization gap*, $\text{acc}[\geq 0.9] - \text{acc}[\leq 0.5]$ (accuracy on near-duplicate test sequences minus accuracy on novel ones), which tracks memorization propensity independently of overall accuracy. A label-concordance analysis records whether each near-duplicate test sequence shares its nearest training neighbour's label.

3.8. From-scratch 1D CNN dose-response

As a deep-model probe we train a from-scratch 1D residual convolutional network (Basset/DeepSEA-style; one-hot input, no k -mer features) on each dataset's training split only, so there is no pretraining confound. We pre-specified, in a timestamped document included in the code release (*results/deep_preregistration.md*), the narrowed claim, the dropout $\times$ weight-decay dose-response grid, and an explicit refutation condition: if the standard-practice regularized cell (dropout 0.3, weight decay 10^{-4}) has a cluster-bootstrap homology-drop CI including 0 on *both* datasets, the strong “any expressive learner is inflated” reading is refuted and reported as a limitation. Models are deterministic (seed 0), trained with early stopping on a held-out slice, on MPS (CPU fallback); the unregularized-to-convergence cell is a labelled manipulation check.

3.9. Cluster (block) bootstrap

Because near-duplicates violate the independence a sample-wise bootstrap assumes, we also resample whole within-test near-duplicate components (8-mer Jaccard > 0.7 connected components) as blocks. We report whether any significance verdict changes, quantify the design effect via the one-way intraclass correlation (ICC), cross-check the block-bootstrap interval against an analytic Liang–Zeger cluster-robust standard error, and report a combined-source interval that folds the five re-split-seed variance into the test-resampling variance.

3.10. Imbalanced evaluation

Because the balanced datasets are curated, we add a prevalence stress test (random forest, target positive fraction $\pi \in \{0.5, 0.2, 0.1\}$) scored with prevalence-aware metrics: AUPRC (against the prevalence baseline), MCC, balanced accuracy, AUROC, and minority-class precision/recall/ F_1 . The near-duplicate-aware splitter is extended with a per-class realized-fraction check so it preserves the target prevalence.

3.11. Injected-leakage multiclass construction

The only multiclass set (*human_ensembl_regulatory*) is clean, so no naturally leaky multiclass data exist in the suite; we answer the multiclass question by construction, injecting label-carrying near-duplicate train→test copies at fraction $f \in \{0, 0.1, 0.2, 0.4\}$ (on a 20k subsample; the mechanism is scale-independent) and comparing the random forest against logistic regression before and after the near-duplicate-aware split.

3.12. Reproducibility

All classical experiments are deterministic with logged seeds, CPU-only, and free of external services; the deep-model runs are seeded and single-device. The near-duplicate-aware splitter is released as a standalone tool (`homology_split.py`, dependencies `numpy/scipy`). Reported 95% confidence intervals are percentile bootstrap intervals; the resampling unit is the individual test sequence (sample bootstrap) or the within-test near-duplicate component (cluster bootstrap), models are not refit, and seeds are fixed.

4. Results

4.1. Leakage is dataset-specific and clean verdicts survive a length-robust re-measure

Of the seven binary datasets, only two—*human_enhancers_ensembl* and *human_nontata_promoters*—exceed a 10% near-duplicate test fraction at Jaccard 0.7 (0.384 and 0.406 respectively); the remaining five are near zero, and the three-class *human_ensembl_regulatory* set is likewise clean (0.005; Table 2). Because a k -mer Jaccard is length-capped, we re-measured at full scale with the length-robust containment index. Containment reveals substantially more leakage on *human_enhancers_ensembl* (0.384 → 0.845; its 2 bp minimum length caps the length-pair Jaccard at ≈ 0.004), and it flags one dataset the length-blind metric misses: *demo_coding_vs_intergenic_seqs*, clean under Jaccard (0.078), crosses the threshold under full-scale containment (0.131), so we relabel it *borderline*. The remaining four clean datasets stay well below the threshold under both metrics (largest full-scale containment 0.068, *human_ocr_ensembl*). Two datasets are thus clearly leaky, one is borderline (containment-only), and four are clean under both metrics; the containment result is itself a demonstration that a length-blind

similarity can under-report leakage on variable-length data, so both metrics should be reported.

4.2. Correcting the split lowers accuracy only on leaky datasets

Under the near-duplicate-aware re-split, the best model's accuracy drops sharply on the leaky datasets—by 16.5 points for the random forest on *human_enhancers_ensembl* (0.8596 → 0.6951 ± 0.0019 ; combined-source 95% CI [15.4, 17.4], folding the test-set and five-seed re-split variance) and 11.8 points on *human_nontata_promoters* (0.9284 → 0.8101 ± 0.0055 ; combined-source 95% CI [9.7, 13.9], centered on the five-seed-mean drop)—whereas every clean-dataset delta is null. All five non-leaky deltas include zero for the memorizing random forest as well as the linear models (the largest, *drosophila_enhancers_stark* RF, is +0.024 with 95% CI [−0.008, 0.058]), so the positive control holds for the model that carries the effect. The negative control confirms causation: a random re-split of the same data, at matched size and balance, leaves the leaky-dataset accuracies essentially unchanged (best-model deltas −0.002 and −0.001, both CIs including zero). Only removing near-duplicates, not re-splitting per se, lowers the score (Fig. 1). The corrected accuracy is stable across five re-split seeds and, independently, across five random-forest training seeds (0.698 ± 0.002 on *human_enhancers_ensembl*, 0.821 ± 0.003 on *human_nontata_promoters*), with the random forest demoted on every seed—so the effect is neither a fortunate partition nor model stochasticity.

4.3. Near-duplicate leakage reorders the model ranking on one dataset

The consequence for method comparison is the central result, and it separates the two leaky datasets. On *human_enhancers_ensembl* the best model *changes*: the default random forest, ranked first on the standard split (accuracy 0.860), falls to last after correction (accuracy 0.696), and *LinearSVC* becomes best (0.798; corrected order *LinearSVC* > *LR* > *HGB* > *RF*; Fig. 2). This flip is airtight along every axis we tested. (i) It holds under all three metrics—the random forest is demoted to rank 4 under accuracy, AUROC (0.805 vs *LinearSVC* 0.870 corrected), and F_1 (0.635 vs 0.796)—so it is not a threshold artifact. (ii) The bootstrap probability that the random forest ranks first goes from $P = 1.0$ on the standard split to $P = 0.0$ after correction, with *LinearSVC* at $P = 1.0$ (both sample- and cluster-bootstrap). (iii) It reproduces under a chromosome-holdout control (§4.8). On *human_nontata_promoters* the picture is weaker. Under accuracy at threshold 0.5 the random forest is demoted from rank 1 to rank 3 (corrected *LR* 0.829 > *HGB* 0.827 > *RF* 0.820 > *LinearSVC* 0.808; these four accuracies are the single seed-0 partition, used for an apples-to-apples within-split comparison, and the forest's 0.820 here is the seed-0 value of the same quantity whose five-seed mean, 0.810, gives the headline drop), but this is a statistical dead heat—the corrected accuracy CIs of the leaders overlap (*RF* [0.812, 0.828] vs *LR* [0.821, 0.837])—and it does not survive a threshold-free metric: under both AUROC and F_1 the random forest

remains rank 1 (corrected AUROC 0.929, the highest of the four). The bootstrap winner probability agrees: after correction no model dominates (LR 0.63, HGB 0.31, RF 0.06, cluster-bootstrap). We therefore claim the ranking-change result cleanly for *human_enhancers_ensembl*, as an existence proof, and only partially for *human_nontata_promoters*. Per-panel rank correlations (Fig. 2) are consistent but, with only four models, Kendall τ is too coarse to be a primary statistic—a fall from first to last flips only its own pairs, leaving $\tau \approx 0$ despite being the most consequential change in the study.

4.4. The mechanism is memorization of label-carrying near-duplicates

Binning test sequences by similarity to training reveals the mechanism directly (Fig. 3). On *human_enhancers_ensembl*, the default random forest scores 1.000 on near-duplicate test sequences (≥ 0.9 similarity) but only 0.770 on novel ones (< 0.5)—a graded gap of +0.230, versus ≈ 0.035 for the other models—and on the novel sequences the random forest (0.770) is actually worse than the linear SVM (0.782). Memorization pays because near-duplicates carry the label: at ≥ 0.9 similarity, 99.99% of near-duplicate test sequences share their nearest training neighbour's label on *human_enhancers_ensembl* ($n = 11,774$) and 99.95% on *human_nontata_promoters* ($n = 2,037$), so recognizing a near-copy is correct essentially for free. On *human_enhancers_ensembl* the inflation is confined to the memorizing model: the random-forest drop is significant while the logistic-regression drop is null (+0.0004, 95% CI $[-0.006, +0.007]$). Clean datasets have near-empty high-similarity bins, leaving nothing to memorize.

4.5. The effect is a property of the default unregularized random forest

Regularizing the random forest dissolves the entire phenomenon monotonically. At the default (`min_samples_leaf=1`), the forest scores exactly 1.000 on the ≥ 0.9 -similarity bin (pure leaves form around individual near-duplicates), with the large drop (0.164 on *human_enhancers_ensembl*, 0.108 on *human_nontata_promoters*), the large graded gap (0.230/0.121), and rank 1. Increasing `min_samples_leaf` to 500 drives the homology drop to ≈ 0 (0.164 $\rightarrow$ -0.001; 0.108 $\rightarrow$ 0.000), collapses the graded gap (0.230 $\rightarrow$ 0.020; 0.121 $\rightarrow$ 0.061), and lowers accuracy on the ≥ 0.9 bin from 1.000 to ~ 0.71 –0.82; capping `max_depth` drives the drop to ≈ 0 likewise. Crucially, the “random forest wins on the leaky split” phenomenon itself requires the unregularized forest: on the standard split the forest holds rank 1 only near the default and has fallen to rank 4 by `min_samples_leaf` ≥ 20 on both datasets. Either intervention—heavy regularization or a near-duplicate-aware split—reveals the truth; the leakage inflation and demotion are a property of the default high-capacity forest practitioners deploy (answering R3.2 and the “untuned-defaults” concern).

4.6. Deep models confirm the mechanism

A from-scratch 1D CNN, trained only on each training split, shows the same inflation, so the effect is not a classical-model artifact. The pre-registered standard-practice regularized reference cell (dropout 0.3, weight decay 10^{-4}) drops on both datasets—+0.043 on *human_nontata_promoters* (cluster-bootstrap CI $[0.025, 0.061]$) and +0.014 on *human_enhancers_ensembl* ($[0.007, 0.021]$), both excluding zero—so the pre-registered refutation condition is not triggered. Nineteen of the 20 cells drop with cluster-bootstrap CIs excluding zero (all nine *human_enhancers_ensembl* grid cells, eight of nine *human_nontata_promoters* grid cells, and both manipulation checks); the sole exception is one *human_nontata_promoters* cell (dropout 0.6, weight decay 10^{-4}) where heavy dropout under-fits, leaving no inflation to lose (accuracy rises 0.020 after correction). Per-cell single-run variance is non-trivial, so we read the mechanism from the grid-level contrast—regularized reference versus unregularized manipulation check—rather than from individual cells. The unregularized-to-convergence manipulation-check cell memorizes hardest, mirroring the unregularized random forest: on *human_enhancers_ensembl* its graded gap is +0.222 (accuracy 0.974 on the ≥ 0.9 bin vs 0.752 novel) with drop +0.076 $[0.068, 0.084]$. Drop magnitude tracks regularization, consistent with “memorization propensity, not capacity” (answering R1.1a/R2.a3/R3.1(i)).

4.7. Alignment-based validation defuses the k -mer-circularity concern

Re-clustering by MMseqs2 alignment identity, feeding the same whole-cluster splitter, reproduces the *human_enhancers_ensembl* effect metric-independently: the random-forest drop is 0.164 under 8-mer Jaccard and 0.162 under alignment identity, the other models stay near zero under both, and the random forest is still demoted to rank 4. On *human_nontata_promoters* the drop is milder under alignment (0.108 $\rightarrow$ 0.064; corrected random-forest accuracy 0.82 $\rightarrow$ 0.864), i.e. partly k -mer-specific—reinforcing its partial status. Because MMseqs2 is k -mer-seeded we claim “alignment-scored,” not “ k -mer-independent” (answering R2.a2/R1.3/R3.4).

4.8. A chromosome-holdout control

Under the genomics-standard chromosome-holdout split, the marquee demotion reproduces on *human_enhancers_ensembl*: the random forest falls from rank 1 (0.8596) to rank 4 (0.7069), giving the identical corrected order (LinearSVC>LR>HGB>RF) as the near-duplicate-aware split. On *human_nontata_promoters* it does not: the random forest stays rank 1 (0.9284 $\rightarrow$ 0.8199; order RF>HGB>LR>LinearSVC), a third independent signal (with the AUROC/ F_1 reversal and the novel-only $\tau = -0.67$) that its demotion is the cautionary partial case. The two controls agree on the drop magnitude (chromosome-holdout corrected 0.707/0.820 vs near-duplicate-aware 0.696/0.810), validating the near-duplicate-aware split as a good proxy for the

deployment control; residual near-duplicate leakage surviving the chromosome split is 0.8% and 0.02%, so it is standard but not leakage-free.

4.9. The cluster (block) bootstrap changes no verdict

Resampling whole within-test near-duplicate components as blocks (the correct tool for correlated near-duplicates, per R3.3) widens the CIs by 1.0–1.5 $\times$ but flips no significance verdict: the random-forest drop still excludes zero on both datasets (*human_enhancers_ensembl* [0.156, 0.172]; *human_nontata_promoters* [0.093, 0.126]) and the *human_enhancers_ensembl* logistic-regression drop stays null. The design effect is small because the correlated near-duplicates are test-to-*train*, not test-to-test—within-test clustering is only 1.06–1.19 sequences per component—and a sample-wise bootstrap of a fixed trained model's test correctness is nearly unbiased in that regime. The intraclass correlation is high (0.91–0.99) yet the design effect is small (1.06–1.17); the analytic cluster-robust interval agrees with the block bootstrap, and a combined-source interval that additionally folds in the five re-split-seed variance still excludes zero for both leaky random-forest drops (*human_enhancers_ensembl* [0.154, 0.174]; *human_nontata_promoters* [0.088, 0.129]), while every clean and three-class delta remains null. Because we report roughly a dozen leaky-versus-clean delta intervals, we also applied a Benjamini–Hochberg correction at $q = 0.05$: it leaves the two leaky random-forest drops and the *human_nontata_promoters* logistic-regression drop significant and no clean or three-class delta significant, so multiplicity changes no verdict.

4.10. Robustness to metric geometry

The findings are robust to how similarity and features are defined. The length-robust containment index reveals more leakage on *human_enhancers_ensembl* (0.845) and flags *demo_coding* as borderline (0.131), while the other four clean datasets stay below the threshold (§4.1). Canonical (reverse-complement-collapsed) k -mer features keep the demotion—random forest 1 $\rightarrow$ 4, drop 0.153 on *human_enhancers_ensembl*; 1 $\rightarrow$ 3, drop 0.097 on *human_nontata_promoters*—so it is genuine memorization, not forward-strand overfitting; canonicalization barely changes the leakage fraction (*human_nontata_promoters* 0.406 $\rightarrow$ 0.416). The drop is not a GC covariate-shift artifact: restricting the corrected test set to GC-in-distribution sequences gives *lower* accuracy (0.787 nontata, 0.631 ensembl), not the inflated original. Finally, single-linkage cohesion explains the ensembl-vs-nontata asymmetry: *human_enhancers_ensembl* clusters are 99.6% pairwise with largest-cluster cohesion 0.96 (clean discrete $\sim 2\times$ locus duplication—correction is a clean fix), whereas *human_nontata_promoters*'s 420-member largest cluster has cohesion 0.083 (24% pairwise), i.e. heavy transitive chaining that over-removes genuine promoter/gene-family signal—the biological reason its demotion is partial. The leaked near-duplicates are not low-complexity or repeat-driven: their dustmasker-masked fraction (0.033 on *human_nontata_promoters*, 0.053 on *human_enhancers_ensembl*) is low and comparable to

that of novel sequences (0.106 and 0.043 respectively)—below on *human_nontata_promoters* and within a point on *human_enhancers_ensembl*—so the near-duplication is genuine discrete locus duplication rather than a transposable-element or microsatellite artifact, and its removal is a legitimate correction.

4.11. Imbalanced setting

Under prevalence imbalance the leakage inflation is revealed by prevalence-aware metrics but masked by accuracy. On *human_enhancers_ensembl* at $\pi = 0.2$, correcting the split lowers AUPRC 0.622 $\rightarrow$ 0.477, MCC 0.422 $\rightarrow$ 0.182, and minority recall 0.258 $\rightarrow$ 0.070, while accuracy barely moves (0.844 $\rightarrow$ 0.808); *human_nontata_promoters* shows the same pattern. Accuracy at threshold 0.5 is thus the wrong metric under imbalance, and the appropriate metrics show the inflation even more starkly. The prevalence-aware splitter preserves the target prevalence (realized positive fraction 0.5001/0.2007/0.1001 $\approx$ target), so it handles imbalance without skewing the minority class (answering R2.a1's second half).

4.12. Injected-leakage multiclass

Because the suite ships no naturally leaky multiclass set, we inject controlled near-duplicate contamination into the clean three-class *human_ensembl_regulatory* set. The random-forest drop grows monotonically with the injected fraction f (+0.0004 at $f=0$; +0.118 at 0.1; +0.179 at 0.2; +0.258 at 0.4), while logistic regression stays nearly flat (≤ 0.066), and the corrected accuracy is stable (~ 0.58) regardless of f because the split removes the injected leakage. The mechanism therefore generalizes to the multiclass setting, appears only when $f > 0$, scales with f , and hits only the memorizer (answering R2.a3's multiclass half by construction).

5. Discussion

Reported performance on some popular genomic benchmarks overstates generalization and—more consequentially—can mislead about which method is best. Anyone using these suites to rank models risks selecting a method for its propensity to memorize near-duplicates rather than to generalize. The controlled variable behind this is memorization propensity (equivalently, effective regularization), not model capacity: the paper's own result—the nominally higher-capacity HistGradientBoosting is barely inflated while the default random forest is heavily inflated, and heavy regularization removes both the memorization and the apparent lead—contradicts a capacity account and supports a regularization one.

We are deliberate about scope. *human_enhancers_ensembl* is an existence proof: the ranking change is airtight on every axis (accuracy, AUROC, F_1 , a bootstrap winner probability, a chromosome-holdout control, and an alignment-scored re-measure). *human_nontata_promoters* is a cautionary partial case: five independent signals (CI overlap, AUROC/ F_1 reversal, chromosome control, novel-only $\tau = -0.67$, and a milder alignment drop) show its demotion is accuracy-only, and its single-linkage clusters chain heavily (0.083 cohesion), so

aggressive de-duplication over-removes genuine promoter/gene-family signal and the random forest retains a real edge on novel sequences. The general reading is thus that near-duplicate leakage can reorder rankings *when a memorization-prone model is in the comparison set*—demonstrated cleanly $n=1$ and partially on a second.

Because the effect is dataset-specific, the contribution is constructive: not “these benchmarks are broken” but “here is which datasets to trust, and a tool to certify any new one.” We package this as a per-dataset report card (Table 2) and a dependency-free splitter. We are careful with the report card’s green verdicts, which carry a heavier evidential burden than red ones: a “clean” entry means only *no forward-strand near-duplicate leakage detected* (subject to the disclosed length-pair cap), not certified homology-free. A practical caution concerns scale: subsampling can mask leakage (*human_enhancers_ensembl* appears almost clean at 20k sequences yet is heavily leaky at full scale), so audits must be run at full dataset scale.

Our study has limits. The pretrained foundation models often expected in such a comparison—DNABERT-2 (Zhou et al., 2023), HyenaDNA (Nguyen et al., 2023), and the Nucleotide Transformer (Dalla-Torre et al., 2024)—are both GPU-gated and, more fundamentally, contaminated by construction. The two leaky datasets’ sequences are extracted from the human reference genome—we recovered their hg38 chromosomal coordinates to build the chromosome-holdout control (§4.8)—so *all* of these test sequences lie within HyenaDNA’s whole-genome (hg38) pretraining corpus and within the human component of DNABERT-2’s and the Nucleotide Transformer’s multispecies corpora; the pretraining-overlap is thus $\approx 100\%$, not a quantity we must estimate. Fine-tuning a model that has already seen the test sequences during pretraining opens a second leakage channel that no train/test re-split can close, so any corrected-split drop it produced would be a confounded lower bound rather than split-effect evidence. We therefore scope these models out and name—and here quantify—the confound rather than absorb it. Whether the foundation models practitioners actually deploy inherit this inflation is a distinct and important question that we leave *genuinely open*: answering it requires an evaluation protocol that accounts for pretraining overlap, itself an instance of the leakage problem we study, and our from-scratch CNN establishes only that the mechanism reaches a trained neural network with no pretraining confound. Containment and alignment identity give lower bounds on leakage, and the chromosome-holdout control leaves a small residual. Natural extensions are a pretraining-overlap-aware foundation-model evaluation, additional suites, and additional biological domains (protein, long-range regulatory).

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest

None declared.

Negative control: only the near-duplicate-aware split removes the inflation

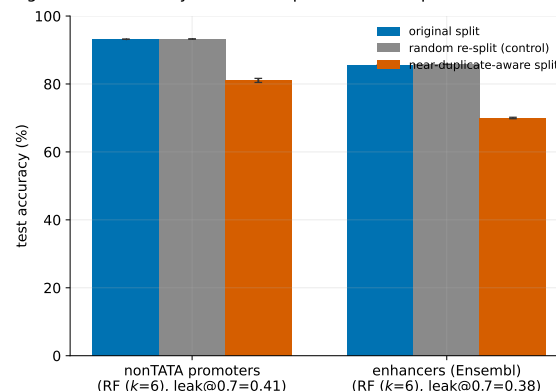


Figure 1 Original split versus random re-split (negative control) versus near-duplicate-aware re-split, best model per dataset. Only the near-duplicate-aware split lowers accuracy, and only on leaky datasets. The y -axis starts at 0; best-model accuracies span roughly 70–93%.

Data Availability

All code required to reproduce the analyses, along with the report card, figures, and the near-duplicate-aware splitter tool, is available at <https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks>. The Genomic Benchmarks datasets analysed here are publicly available and were obtained via the genomic-benchmarks package (Grešová et al., 2023); they are not redistributed in our repository.

References

- A. Bushuiev, R. Bushuiev, J. Sedlar, T. Pluskal, J. Damborsky, S. Mazurenko, and J. Sivic. Revealing data leakage in protein interaction benchmarks. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2404.10457.
- H. Dalla-Torre, L. Gonzalez, J. Mendoza-Revilla, N. Lopez Carranza, A. H. Grzywaczewski, F. Oteri, C. Dallago, E. Trop, B. P. de Almeida, H. Sirelkhatim, et al. The nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 2024. doi: 10.1038/s41592-024-02523-z.
- K. Grešová, V. Martinek, D. Čechák, P. Šimeček, and P. Alexiou. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24 (1):25, 2023. doi: 10.1186/s12863-023-01123-8.
- A. Huang, I. S. Knight, and S. Naprienko. Data leakage and redundancy in the LIT-PCBA benchmark. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2507.21404.
- S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9): 100804, 2023. doi: 10.1016/j.patter.2023.100804.
- D. Kovtun, M. Akdel, A. Goncarenko, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, E. Rossi, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein, and L. Naef. PINDER: The protein interaction dataset and evaluation resource. *bioRxiv preprint*, 2024. doi: 10.1101/2024.07.17.603980.

Table 1. Datasets in the Genomic Benchmarks binary suite (and the optional three-class set). Lengths are the median (with min–max range where variable); class balance is the positive-class fraction (binary). The 2 bp minimum in *human_enhancers_ensembl* caps its length-pair *k*-mer Jaccard at ≈ 0.004 , so its Jaccard leak fraction is a severe lower bound (see the containment column of Table 2).

Dataset	<i>n</i> (full)	Length (bp)	Class balance	Test fraction
human_nontata_promoters	36,131	251	0.544	0.25
human_enhancers_ensembl	154,842	269 (2–573)	0.500	0.20
human_ocr_ensembl	174,756	315 (71–593)	0.500	0.20
human_enhancers_cohn	27,791	500	0.500	0.25
demo_coding_vs_intergenomic_seqs	100,000	200	0.500	0.25
demo_human_or_worm	100,000	200	0.500	0.25
drosophila_enhancers_stark	6,914	2142 (236–3237)	0.500	0.25
human_ensembl_regulatory (3-class)	289,061	401 (89–802)	3 classes	0.20

Table 2. Near-duplicate leakage report card. **Verdict** is LEAKY if the full-scale near-duplicate test fraction exceeds 0.1 at Jaccard 0.7; a “clean” verdict means only *no forward-strand near-duplicate leakage detected*, subject to the disclosed length-pair cap. “leak@0.7 (Jac / con)” gives the leak fraction under the 8-mer Jaccard and the length-robust containment index. “Acc. lost” is the best-model accuracy change under the near-duplicate-aware re-split (positive = accuracy lost; accuracy fraction). “RF rank” is the random-forest rank before→after correction. “Reorders under chrom.-holdout?” reports whether the best-model change reproduces under the chromosome-holdout control (measured for the two leaky datasets only). All leak fractions are full-scale. [†]*demo_coding_vs_intergenomic_seqs* is clean under 8-mer Jaccard (0.078) but its full-scale containment leak fraction (0.131) exceeds 0.1—a borderline case the length-robust metric flags and the length-blind one misses.

Dataset	<i>n</i> (full)	leak@0.7 (Jac)	leak@0.7 (con)	Verdict	Acc. lost	RF rank	Reorders under chrom.-holdout?
human_nontata_promoters	36,131	0.406	0.445	LEAKY	+0.118	1→3	no (RF stays rank 1)
human_enhancers_ensembl	154,842	0.384	0.845	LEAKY	+0.165	1→4	yes (RF → rank 4)
demo_coding_vs_intergenomic_seqs	100,000	0.078	0.131	borderline [†]	−0.001	4→4	n/a
human_ocr_ensembl	174,756	0.010	0.068	clean	+0.004	4→4	n/a
demo_human_or_worm	100,000	0.012	0.042	clean	−0.004	4→4	n/a
drosophila_enhancers_stark	6,914	0.016	0.033	clean	+0.023	2→3	n/a
human_enhancers_cohn	27,791	0.001	0.005	clean	−0.004	4→4	n/a
human_ensembl_regulatory (3-class)	289,061	0.005	–	clean	+0.001	n/a	n/a

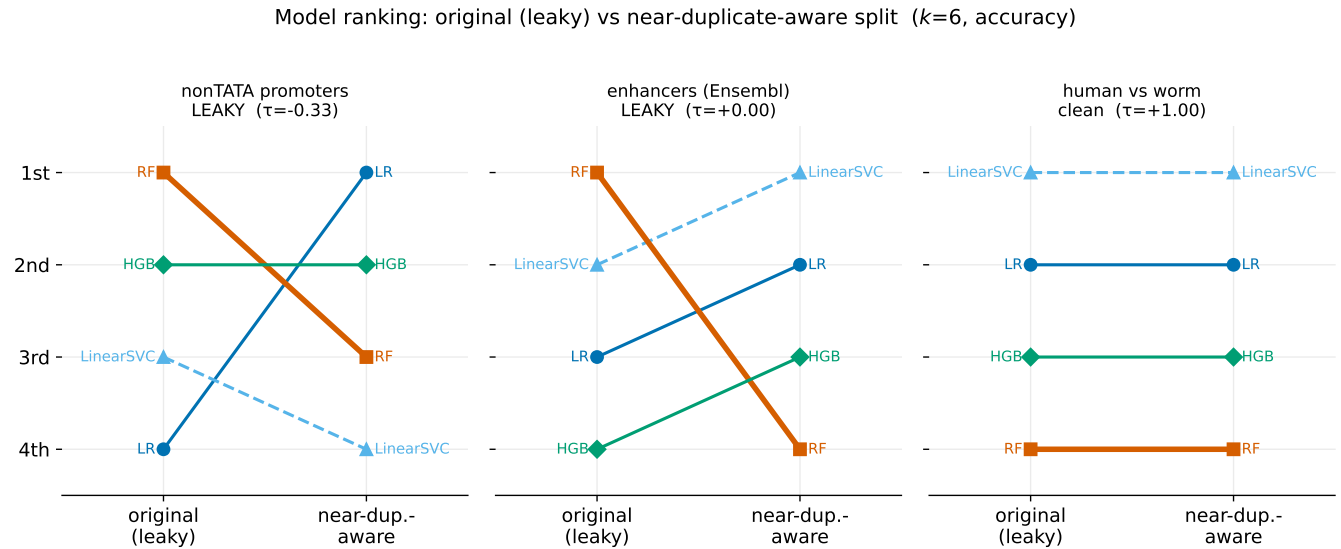


Figure 2 Model rankings on the original (leaky) split versus the near-duplicate-aware split (*k*=6, accuracy). The random forest is dethroned on *human_enhancers_ensembl* (left) and, under accuracy only, on *human_nontata_promoters* (middle), but rankings are stable on a clean control (right). Per-panel τ is the rank correlation; on *human_enhancers_ensembl* the random forest’s move to last flips only its own pairs, giving $\tau \approx 0$ despite the dramatic demotion—illustrating why τ is too coarse to serve as a primary statistic with only four models (§4.3).

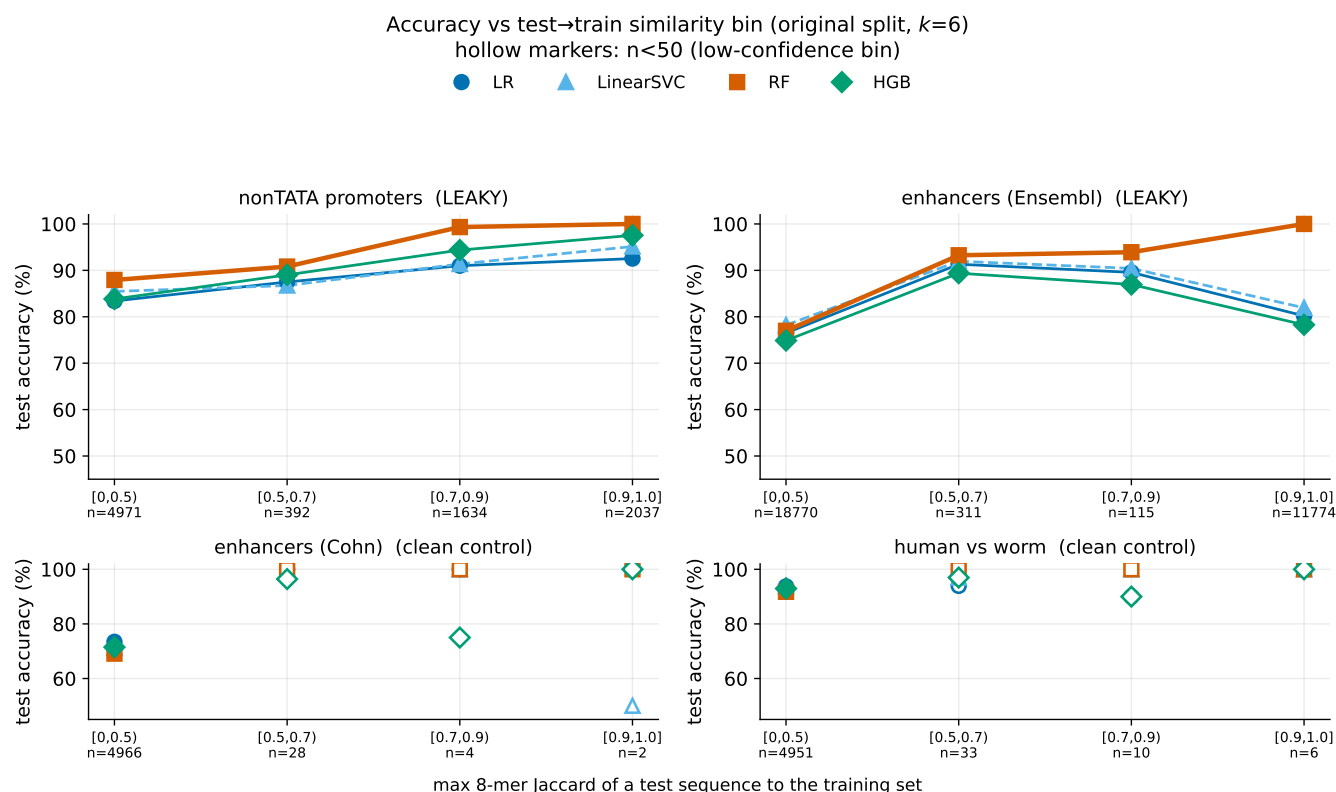


Figure 3 Accuracy as a function of the test-to-train similarity bin (original split, $k=6$). On leaky datasets the default random forest is near-perfect on high-similarity (near-duplicate) test sequences and falls on dissimilar ones; clean controls have near-empty high-similarity bins (hollow markers: $n < 50$).

- W. Li and A. Godzik. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics*, 22(13):1658–1659, 2006. doi: 10.1093/bioinformatics/btl158.
- E. Nguyen, M. Poli, M. Faizi, A. Thomas, C. Birch-Sykes, M. Wornow, A. Patel, C. Rabideau, S. Massaroli, Y. Bengio, S. Ermon, S. A. Baccus, and C. Ré. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. doi: 10.48550/arXiv.2306.15794.
- A. M. Rafi, B. Kiyota, N. Yachie, and C. de Boer. Detecting and avoiding homology-based data leakage in genome-trained sequence models. *bioRxiv preprint*, 2025. doi: 10.1101/2025.01.22.634321.
- M. Steinegger and J. Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11):1026–1028, 2017. doi: 10.1038/nbt.3988.
- F. Teufel, M. H. Gíslason, J. J. Almagro Armenteros, A. R. Johansen, O. Winther, and H. Nielsen. GraphPart: homology partitioning for biological sequence analysis. *NAR Genomics and Bioinformatics*, 5(4):lqad088, 2023. doi: 10.1093/nargab/lqad088.
- Z. Zhou, Y. Ji, W. Li, P. Dutta, R. Davuluri, and H. Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. *arXiv preprint*, 2023. doi: 10.48550/arXiv.2306.15006.